

MERN + AI Job-Ready Bootcamp – 4 Month Curriculum

This program is designed to take students from beginner level to job-ready Fullstack developer with real-world projects, automation, APIs and AI exposure.

Week	Lecture	Topics Covered	Practical / Outcome
Week 1	L1	How the Internet Works, Client vs Server, HTTP Basics	Explain request-response cycle
	L2	HTML Structure, Semantic Tags & SEO Basics	Build structured HTML page
	L3	Forms & Input Types + AI for HTML Boilerplate	Create responsive form using AI
Week 2	L4	CSS Fundamentals, Box Model & Selectors	Style static webpage
	L5	Flexbox & Responsive Design Principles	Build responsive layout
	L6	Intro to Tailwind CSS + AI Debugging CSS	Design modern landing page
Week 3	L7	JavaScript Basics – Variables, Data Types, Operators	Write simple JS programs
	L8	Functions, Arrays, Objects & ES6 Features	Array & object challenges
	L9	Async JavaScript – Promises, Async/Await, Fetch API	Build API-based mini app
Week 4	L10	Git & GitHub Fundamentals, Branching Strategy	Push project to GitHub
	L11	Clean Code, Folder Structure & Documentation	Refactor messy project
	L12	AI for Developers – Prompt Engineering Basics	Use AI for debugging & commits
Week 5	L13	React Introduction – Components, JSX, Props	Build reusable components
	L14	State, useEffect & Event Handling	Create interactive UI
	L15	React Router & Multi-Page Navigation	Build multi-page React app
Week 6	L16	Forms in React & Controlled Components	Build form with validation
	L17	API Integration with Axios + Interceptors	Connect frontend to API
	L18	Reusable Component Design + AI Refactoring	Optimize React structure
Week 7	L19	Context API & State Management	Implement global state
	L20	Authentication Flow & Protected Routes	Secure frontend routes
	L21	Frontend Error Handling & UI Polish with AI	Improve UX & accessibility
Week 8	L22	Frontend Production Project – Setup	Initialize dashboard project
	L23	Frontend Production Project – CRUD + Auth	Build full CRUD system
	L24	Frontend Project Review & Optimization	Finalize production-ready frontend

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Week 9	L25	Node.js Fundamentals & Express Setup	Create basic server
	L26	Middleware & REST API Design Principles	Build structured REST API
	L27	Service Layer Architecture & Clean Backend Structure	Implement scalable folder structure
Week 10	L28	Database Design – MongoDB OR PostgreSQL + Prisma	Design relational schema
	L29	CRUD Operations & Relationships	Implement relational CRUD
	L30	AI for Schema Generation & Query Optimization	Improve database performance
Week 11	L31	Authentication – JWT, Access vs Refresh Tokens	Implement secure login system
	L32	Password Hashing & Role-Based Access Control	Secure role-based routes
	L33	Backend Security Best Practices	Harden API endpoints
Week 12	L34	Backend Production Project – Setup	Initialize e-commerce/booking backend
	L35	Backend Production Project – Advanced Features	Implement business logic
	L36	Backend Project Testing & Optimization	Finalize production backend
Week 13	L37	AI for Developers – Code Generation & Refactoring	Use AI for structured coding
	L38	AI for Testing, Debugging & Documentation	Generate test cases & docs
	L39	Responsible AI Usage in Development	Apply AI safely in workflow
Week 14	L40	Integrating OpenAI API – Build Simple Chatbot	Create AI chatbot feature
	L41	AI Blog Generator / Content Tool	Build AI content tool
	L42	AI Code Reviewer Tool	Create AI review assistant
Week 15	L43	Environment Variables & Production Configurations	Secure environment setup
	L44	Deploy Frontend (Vercel) & Backend	Deploy full-stack app
	L45	Intro to CI/CD & Production Best Practices	Understand deployment pipelines
Week 16	L46	Capstone Project Planning & Architecture Design	Finalize project blueprint
	L47	Capstone Development + AI Feature Integration	Build complete full-stack system
	L48	Final Project Review + Career Preparation	Resume, GitHub & mock interview
Bonus	L49	Advanced Backend Architecture Patterns	Implement scalable enterprise structure
	L50	Performance Optimization & Scaling Applications	Optimize production system